

ASSIGNMENT NO.7: Subqueries and Nested Queries

Questions:

Q1) Create a MySQL database named CollegeDB and create two tables: Course and StudentResult.

Course Table

Column Name	Data Type	Constraint
CourseID	INT	PRIMARY KEY
CourseName	VARCHAR(50)	NOT NULL
Fees	DECIMAL(10,2)	NOT NULL

StudentResult Table

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	NOT NULL
CourseID	INT	FOREIGN KEY
City	VARCHAR(30)	
Marks	INT	

Insert at least 4 records into Course and 10 records into StudentResult.

Answer:-

My SQLQUERY:

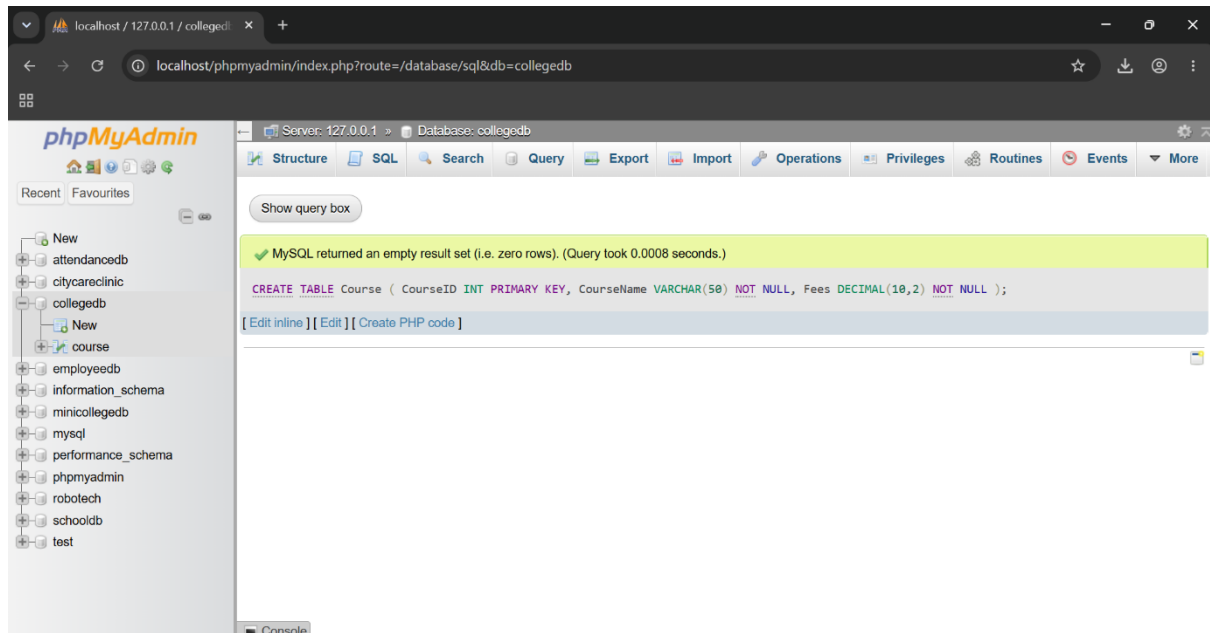
CREATE TABLE Course (

CourseID INT PRIMARY KEY,

CourseName VARCHAR(50) NOT NULL,

Fees DECIMAL(10,2) NOT NULL

);



CREATE TABLE StudentResult (

StudentID INT PRIMARY KEY,

StudentName VARCHAR(50) NOT NULL,

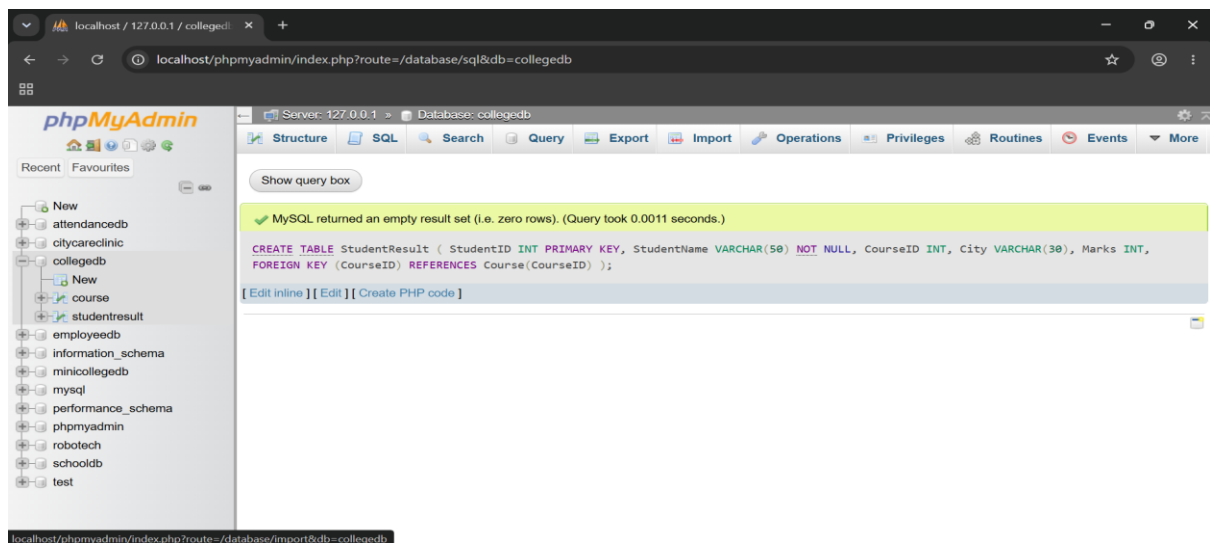
CourseID INT,

City VARCHAR(30),

Marks INT,

FOREIGN KEY (CourseID) REFERENCES Course(CourseID)

);



localhost / 127.0.0.1 / collegedb

localhost/phpmyadmin/index.php?route=/database/sql&db=collegedb

phpMyAdmin

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 - studentresult
- employeeeb
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- phpmyadmin
- robotech
- schooldb
- test

Server: 127.0.0.1 > Database: collegedb

Structure SQL Search Query Export Import Operations Privileges Routines Events More

Show query box

4 rows inserted. (Query took 0.0047 seconds.)

```
INSERT INTO Course (CourseID, CourseName, Fees) VALUES (101, 'BSc IT', 45000.00), (102, 'BCA', 40000.00), (103, 'BSc CS', 42000.00), (104, 'Data Science', 55000.00);
```

[Edit inline] [Edit] [Create PHP code]

Console

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- robotech
- schooldb
- test

Server: 127.0.0.1 > Database: collegedb > Table: Course

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking More

Show query box

Showing rows 0 - 3 (4 total, Query took 0.0014 seconds.)

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	CourseID	CourseName	Fees
☐ Edit ☐ Copy ☐ Delete	101	BSc IT	45000.00
☐ Edit ☐ Copy ☐ Delete	102	BCA	40000.00
☐ Edit ☐ Copy ☐ Delete	103	BSc CS	42000.00
☐ Edit ☐ Copy ☐ Delete	104	Data Science	55000.00

☐ Check all | With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

Console w all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

localhost / 127.0.0.1 / collegedb

localhost/phpmyadmin/index.php?route=/table/sql&db=collegedb&table=Course

phpMyAdmin

Recent Favourites

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- robotech
- schooldb
- test

Server: 127.0.0.1 > Database: collegedb > Table: Course

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking More

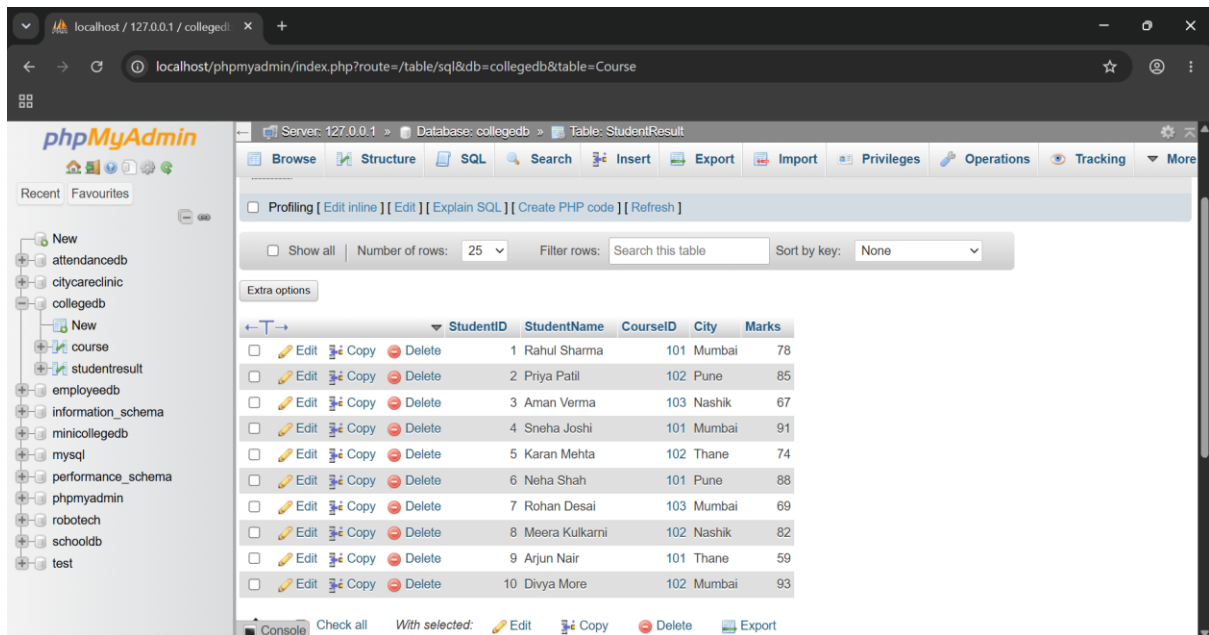
Show query box

10 rows inserted. (Query took 0.0010 seconds.)

```
INSERT INTO StudentResult (StudentID, StudentName, CourseID, City, Marks) VALUES (1, 'Rahul Sharma', 101, 'Mumbai', 78), (2, 'Priya Patil', 102, 'Pune', 85), (3, 'Aman Verma', 103, 'Nashik', 67), (4, 'Sneha Joshi', 101, 'Mumbai', 91), (5, 'Karan Mehta', 102, 'Thane', 74), (6, 'Neha Shah', 101, 'Pune', 88), (7, 'Rohan Desai', 103, 'Mumbai', 69), (8, 'Meera Kulkarni', 102, 'Nashik', 82), (9, 'Arjun Nair', 101, 'Thane', 59), (10, 'Divya More', 102, 'Mumbai', 93);
```

[Edit inline] [Edit] [Create PHP code]

Console



Q2) Write a MySQL query to display students whose marks are greater than the average marks of all students.

The output should contain:

Column
StudentName
Marks

Answer:-

MySQL query

SELECT StudentName, Marks

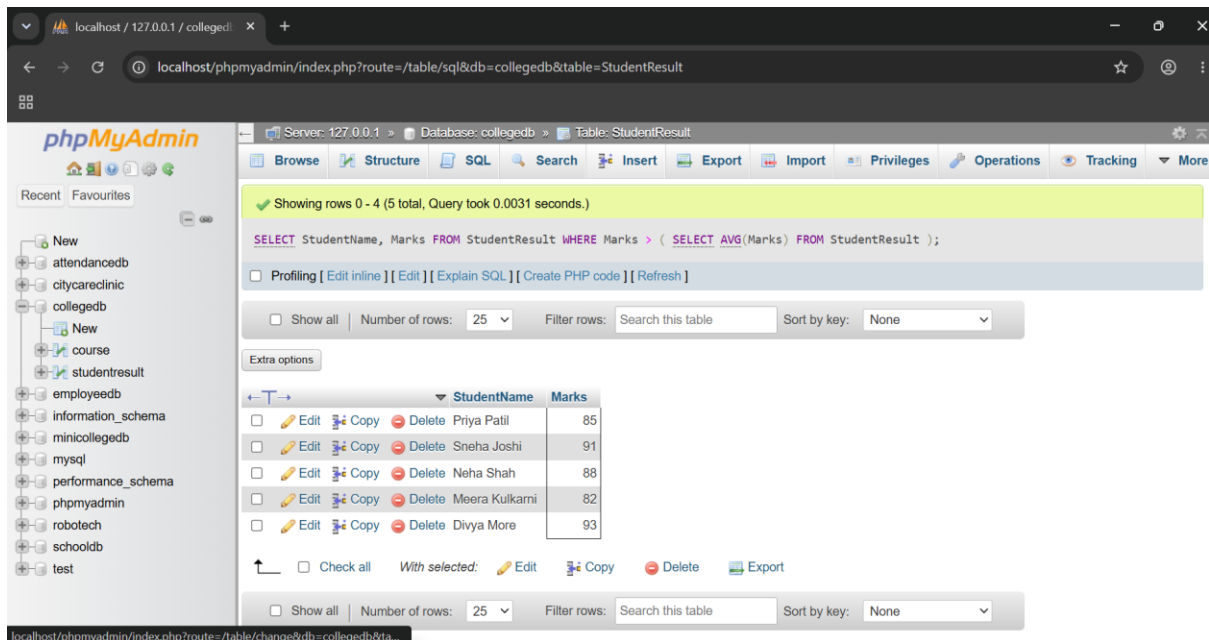
FROM StudentResult

WHERE Marks > (

SELECT AVG(Marks)

FROM StudentResult

);



Q3) Write a MySQL query to display all students enrolled in the BCA course using a subquery.

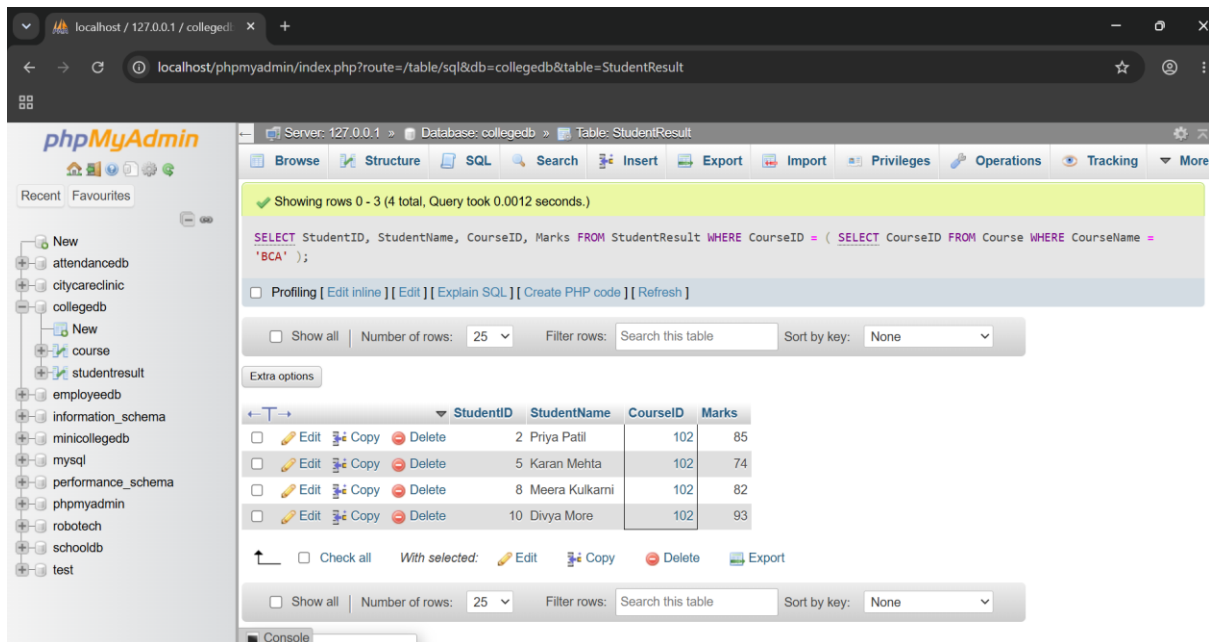
The output should contain:

Column
StudentID
StudentName
Column
CourseID
Marks

Answer:-

MySQL query

```
SELECT StudentID, StudentName, CourseID, Marks
FROM StudentResult
WHERE CourseID = (
    SELECT CourseID
    FROM Course
    WHERE CourseName = 'BCA');
```



Q4) Write a MySQL query to display each student with the overall average marks using a subquery in the SELECT clause.

The output should contain:

Column
StudentName
Marks
AverageMarks

Answer:-

MySQL query

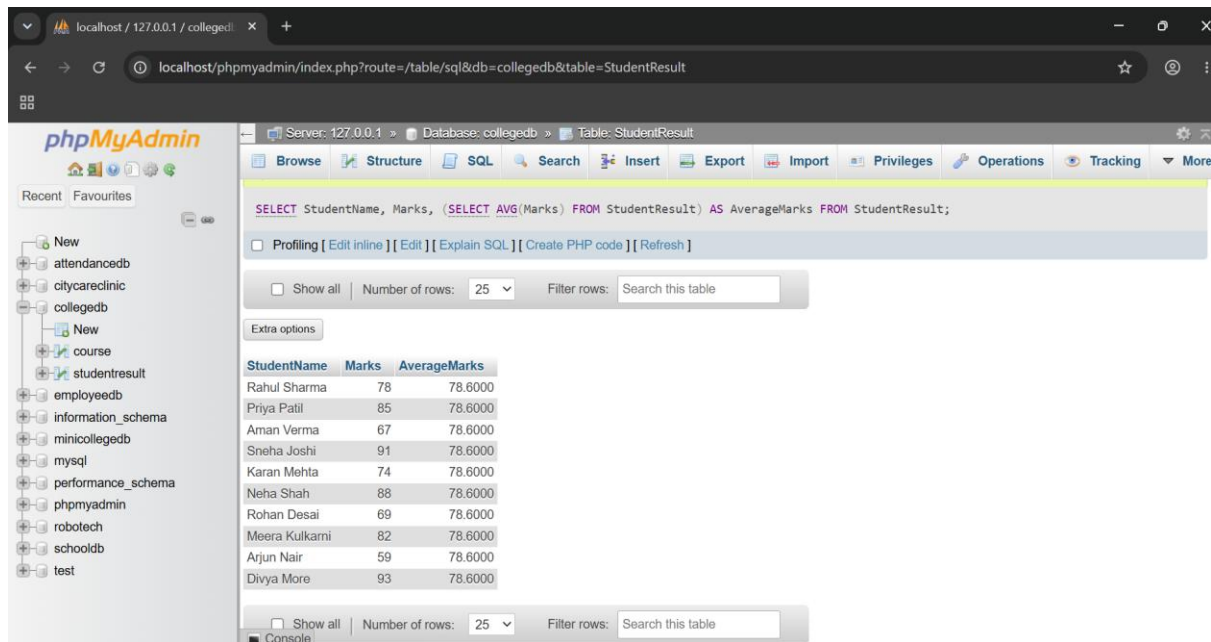
SELECT

StudentName,

Marks,

(SELECT AVG(Marks) FROM StudentResult) AS AverageMarks

FROM StudentResult;



Q5) Write a MySQL query using a correlated subquery to display students who scored more than their own course average.

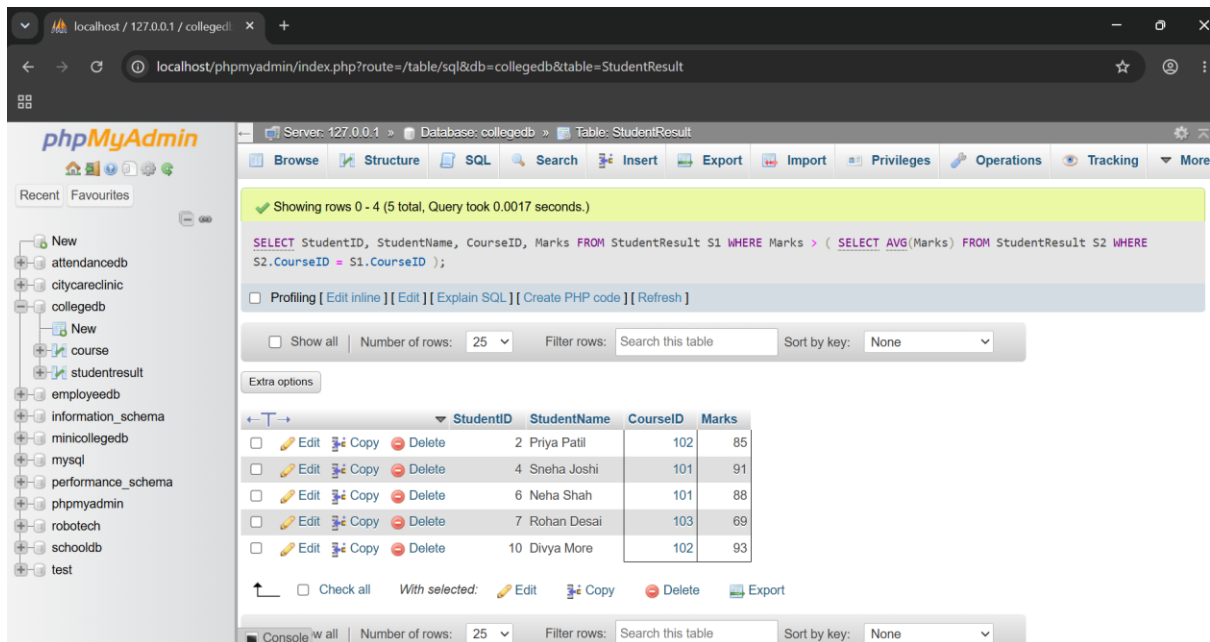
The output should contain:

Column
StudentID
StudentName
CourseID
Marks

Answer:-

MySQL query

```
SELECT StudentID, StudentName, CourseID, Marks
FROM StudentResult S1
WHERE Marks > (
    SELECT AVG(Marks)
    FROM StudentResult S2
    WHERE S2.CourseID = S1.CourseID);
```



Q6) Write MySQL queries to display courses having students and courses having no students.

Use:

Operator
EXISTS
NOT EXISTS

Answer:-

MySQL query

SELECT CourseID, CourseName

FROM Course C

WHERE EXISTS (

SELECT 1

FROM StudentResult S

WHERE S.CourseID = C.CourseID

);

localhost / 127.0.0.1 / collegedb

localhost/phpmyadmin/index.php?route=/table/sql&db=collegedb&table=StudentResult

Server: 127.0.0.1 Database: collegedb Table: Course

Showing rows 0 - 2 (3 total, Query took 0.0015 seconds.)

```
SELECT CourseID, CourseName FROM Course C WHERE EXISTS ( SELECT 1 FROM StudentResult S WHERE S.CourseID = C.CourseID );
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	CourseID	CourseName
☐ Edit ☐ Copy ☐ Delete	101	BSc IT
☐ Edit ☐ Copy ☐ Delete	102	BCA
☐ Edit ☐ Copy ☐ Delete	103	BSc CS

☐ Check all With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Console

SELECT CourseID, CourseName

FROM Course C

WHERE NOT EXISTS (

SELECT 1

FROM StudentResult S

WHERE S.CourseID = C.CourseID

);

localhost / 127.0.0.1 / collegedb

localhost/phpmyadmin/index.php?route=/table/sql&db=collegedb&table=Course

Server: 127.0.0.1 Database: collegedb Table: Course

Showing rows 0 - 0 (1 total, Query took 0.0029 seconds.)

```
SELECT CourseID, CourseName FROM Course C WHERE NOT EXISTS ( SELECT 1 FROM StudentResult S WHERE S.CourseID = C.CourseID );
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

	CourseID	CourseName
☐ Edit ☐ Copy ☐ Delete	104	Data Science

☐ Check all With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

Print ☐ Copy to clipboard ☐ Export ☐ Display chart ☐ Create view

Console